



Reduction of metronidazole in municipal wastewater and protection of activated sludge system using a novel immobilized *Aspergillus tabacinus* LZ-M

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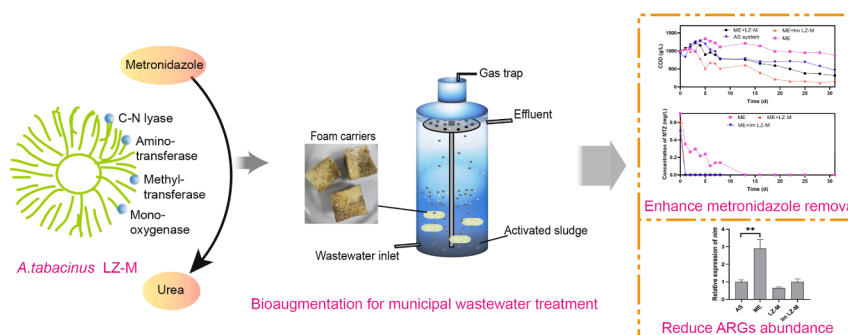
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HIGHLIGHTS

- *A. tabacinus* LZ-M degraded 77.39 % of 5 mg/L metronidazole (MNZ) into urea.
- Enzymes were expressed for —C—N— and —C—C— cleavage and hydroxylation.
- Immobilized LZ-M promoted the 100% removal of MNZ in municipal wastewater.
- Bioaugmentation protected sludge structure and microbial community from MNZ.
- Bioaugmentation reduced the abundance of *nim* resistance genes.

GRAPHICAL ABSTRACT



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ABSTRACT

Metronidazole (MNZ) accumulation inhibits municipal wastewater treatment bio-systems, and an effective solution to augment anaerobic activated sludge (AAS) is required. This research discovered that *Aspergillus tabacinus* LZ-M could degrade 77.39% of MNZ at 5 mg/L. MNZ was metabolized into urea, and the enzymes involved in its degradation were aminotransferase, methyltransferase, monooxygenase, and —C—N— cleavage hydrolase. The strain was immobilized in polyurethane foam and used in AAS for the treatment of MNZ-containing municipal wastewater. The results showed that, using immobilized LZ-M, MNZ was completely removed, and the degradation efficiency of wastewater's chemical oxygen demand (COD) was increased from 11.7% to

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83.31%. The extracellular polymer and ROS levels indicated that MNZ's toxicity on AAS was reduced. Furthermore, bioaugmentation stabilized its microbial community, and decreased MNZ resistance genes. These observations confirm that the immobilized fungi are effective in protecting AAS against antibiotic contamination in the treatment process of municipal wastewater.

1. Introduction

Antibiotics have been confirmed as crucial pharmaceuticals. They have been widely employed for preventing and treating animals' and humans' infectious diseases (Gudda et al., 2020). Global consumption of antibiotics was evaluated at 63,151 tons in 2010 and is projected to be 105,596 ($\pm 3,605$) tons in 20 years (Van Boeckel et al., 2015). Antibiotic-resistant bacteria and antibiotic-resistant genes (ARGs) emerge in the environment, posing an urgent threat to the global environment (Chamusa et al., 2017). Wastewater treatment plants (WWTPs) are major storage sites for antibiotics and ARGs (Gudda et al., 2020). Especially, the municipal wastewater treatment system, even exhibited a higher level of resistance gene diversity and richness than hospital wastewater treatment (Zhang et al., 2021). Metronidazole (MNZ) is commonly used to treat diseases associated with parasites, microaerophilic bacteria, and anaerobic bacteria. It is also added to poultry and fish feed to help eliminate parasites (Pan et al., 2019). Residual amounts of it (0.5 to 10 $\mu\text{g/L}$) have been found in sewage treatment plant effluent (Pan et al., 2019). It inhibits anaerobes' activity in anaerobic activated-sludge (AAS) systems, causing the instability and low efficiencies of municipal wastewater treatment and increasing the spread of ARGs in WWTPs (Czatzkowska et al., 2021). To facilitate antibiotic degradation in municipal wastewater, effective methods must be developed.

Chemical and physical methods in treatment of antibiotic wastewater have the disadvantages of high costs and secondary pollution (Pan et al., 2019). Some advanced oxidation technologies are efficient for the remediation of hospital sewage or re-treatment of wastewater (Calcio Gaudino et al., 2021), whereas they do not apply to the treatment of antibiotics in municipal wastewater, which has a large discharge and high chemical oxygen demand (COD) (Aboudalle et al., 2018). Biodegradation is economical and environmentally friendly. Bioaugmentation of activated sludge systems is often used to promote the degradation of toxic substances in wastewater (Ji et al., 2019; Ma et al., 2016). However, due to the antibacterial properties of metronidazole, no microbes are isolated to cause its decomposition. Bacteria are highly sensitive to antibiotics and need to be enriched and cultured for a long time, which will lead to the risk of transmission of resistance genes. Fungi as effective decomposers in the environment, are capable of transforming and mineralizing extensive organic pollutants based on the activity of nonspecific proteases (e.g., laccase, lignin peroxidase, manganese peroxidase, as well as versatile peroxidase), and showing excellent ability in the degradation of some antibiotics (Čvančarová et al., 2015). For instance, norfloxacin, ofloxacin, and ciprofloxacin fluoroquinolones are degraded by white-rot fungi and *T. versicolor* (Čvančarová et al., 2015). The flumequine is degraded by zygomycete *Cunninghamella elegans* (Williams et al., 2007). Thus fungi have great potential for antibiotic elimination in WWTPs.

When fungi are employed in bioreactors for wastewater treatment, the difficult coexistence of fungi and native microorganisms due to interspecific competition is a problem for efficient bioaugmentation (Hailei et al., 2017). Mycelial pellets of white-rot fungus have been used in activated sludge systems to enhance coking wastewater treatment (Hailei et al., 2017). Whereas mycelial pellets is prone to breakage in composite systems (Espinosa-Ortiz et al., 2016). Few immobilized fungal techniques have been used in bioreactors to improve antibiotic degradation. For example, tetracycline wastewater treatment is performed by whole-cell immobilization of fungi combined with bacteria (Wu et al., 2021). Wheat straw pellets were used as attachment carriers for white-rot fungi *Trametes versicolor* and *Bjerkandera adusta* for the

treatment of pharmaceutical products manufacturing wastewaters (Aydin, 2016). However, fungal immobilization for MNZ degradation has not been reported. Therefore, effective immobilized technologies could be developed to achieve the application of filamentous fungi in activated sludge systems for MNZ degradation.

In this study, *Aspergillus tabacinus* LZ-M degraded MNZ efficiently. To investigate its mechanism and application, the MNZ degradation pathway was analyzed using LC-MS, and the transcriptome and proteome were used to examine the enzymes involved in MNZ degradation. The LZ-M was then immobilized in polyurethane foam and introduced into the AAS system for bioaugmentation to treat MNZ-containing municipal wastewater. MNZ, COD, and gas production changes in reactors were studied. After bioaugmentation, the activated sludge community and reduced resistance genes were studied. The feasibility of fungus-mediated bioaugmentation in the treatment of municipal wastewater containing MNZ was revealed.

2. Materials and methods

2.1. Fungal strain and culture

Metronidazole (MNZ) was provided by Sigma-Aldrich (St. Louis, USA). *Aspergillus tabacinus* LZ-M was separated from soil samples in Gansu Academy of Membrane Science and Technology (36°5'N, 103°88'E), and identified by ITS sequencing analysis. It grew aerobically in conical flasks with a volume of 250 mL, in which 150 mL of Potato Dextrose Broth Medium (PDB, Solarbio, China) was contained and 150 μL of spore suspension was inoculated. The culture was performed for 60 h at 30 °C and 200 rpm, and then mycelia pellets were formed in the flasks.

2.2. Degradation of MNZ by strain LZ-M

In the degradation experiment, 200 g of LZ-M wet pellets (equivalent to 3.1 g/L dry weight) were suspended in 1 L of basal medium (BM) having 1 g/L NaCl, 1 g/L KH_2PO_4 , 1 g/L K_2HPO_4 , and 1 g/L soluble starch at pH 7.0. Adding MNZ into BM to reach a concentration of 5 mg/L. Following that, 250 mL anaerobic bottles imported 100 mL of the fungal suspension and filled with nitrogen for 20 min, and then they were cultured anaerobically at 30 °C for 5 days. Experiment using sterile medium and aerobic culture as controls. For aerobic culture, the bottles were incubated at 30 °C and 200 rpm. The samples were acquired at various periods for MNZ determination. The respective experiment was performed in duplicate.

The effect factors of biodegradation of MNZ by strain LZ-4 under anaerobic conditions were analyzed. To examine the effect of biomass concentration of mycelia pellets, the LZ-M with different biomass concentrations of 0.5–5 g/L (dry weight) were introduced into BM, respectively. To investigate the influence of MNZ's initial concentration, the MNZ concentrations in BM were set from 1 to 50 mg/L, respectively. The 30 mg/L furazolidone (FZD) and 30 mg/L 3,5-dinitrosalicylic acids (DNSA) were introduced into BM to examine the removal of other pharmaceutical compounds by strain LZ-M. The dry weight of fungus LZ-M was 3.1 g/L except if the single factor was changed in accordance with the experimental plan. The bottles were incubated for 5 days at 30 °C. All the experimental procedures were performed in duplicate.

2.3. Bioassay for antibacterial activity of metronidazole metabolites

The growth of *Clostridium perfringens* was monitored to examine the antibacterial activity of MNZ metabolites. *C. perfringens*, which cannot grow in media containing 2 mg/L of MNZ (Rafii et al., 2003), was grown overnight in 5 mL Man, Rogosa, and Sharpe broth (Solarbio Sci & Tech, Beijing, China) at 37 °C. Then the supernatants of MNZ culture medium degraded by LZ-M and not degraded by LZ-M were used to suspend *C. perfringens* in anaerobic tubes to obtain a cell concentration of about 10⁸ cells/mL. The bacterial suspensions were cultured at 37 °C for 24 h. After gradient dilution, 5 µL of the bacterial solution was dropped on the MRS solid culture plates and cultured in an anaerobic incubator at 37 °C for 24 h. The growth of *C. perfringens* was investigated at the end of cultivation.

2.4. Immobilization of *A. tabacinus* LZ-M on polyurethane foam

Polyurethane foam (density 22 kg/m³) was cut into blocks (1 cm*1 cm*1 cm). Thirty pieces of polyurethane foam and 100 mL of PDB medium were introduced into a 500 mL Erlenmeyer flask and inoculated with 300 µL spore suspension of *A. tabacinus* LZ-M. The flask was cultured at 150 rpm and 30 °C for 14 days, and green colonies appeared on the foam. The foam were washed three times with ddH₂O and employed for wastewater treatment.

2.5. Bioreactor operations

The wastewater was obtained from the sewage inlet at the Lanzhou Yanerwan Sewage Treatment Factory in Lanzhou, Gansu, China, and the AAS was obtained from the treatment process's anaerobic tank. The pH of the wastewater reached 7.6. Furthermore, the COD concentration reached 1050 mg/L.

Four small up-flow anaerobic sludge Bed (UASB) reactors were set with a working volume of 1 L, an inner diameter of 9 cm, and a height of 25 cm. The respective reactor was charged with 276.5 mL of activated sludge and 724 mL of sewage, to reach an MLSS concentration of 3.78 g/L. For different treatment, AS reactor was employed as a control system without MNZ; ME reactor was supplied with 1 mg/L MNZ in wastewater; ME+LZ-M reactor was supplied with 1 mg/L MNZ in wastewater, and 100 g of LZ-M mycelial pellets (wet weight) was inoculated for bioaugmentation; ME+ImLZ-M reactor was supplied with 1 mg/L MNZ in wastewater, and 30 block foams attached with strain LZ-M were introduced to the reactor for immobilized bioaugmentation. To exclude the effect of the foam, an equal amount of sterile polyurethane foam was introduced into the AS, ME, and ME + LZ-M reactor, respectively. The four reactors were filled with nitrogen for 20 min and operated for 31 days in an incubator of 30 °C. The effluent in a volume of 200 mL was discharged daily and then replaced with new wastewater. The effluent and activated sludge were acquired at different periods for analysis.

2.6. Transcriptome and proteome sequencing of *A. tabacinus* LZ-M and 16S rDNA high-throughput sequencing of activated sludge

The wet mycelia pellets of LZ-M having a mass of 20 g were transferred to 100 mL BM with 5 mg/L MNZ or without MNZ, and fungal suspensions were incubated under anaerobic conditions at 30 °C for 3.5 h. To prepare the sample for proteome sequencing, mycelia pellets were cultured for 3 days. Activated sludge was collected from the reactors. The obtained mycelia pellets or activated sludge were sent to Shanghai Majorbio Bio-pharm Technology Co., Ltd. (Shanghai, China) for sequencing analysis. The transcriptome analysis was conducted using the TruSeqTM RNA sample preparation Kit (Illumina, San Diego, CA) and the Illumina HiSeq xten/NovaSeq 6000 sequencer for paired-end sequencing. Proteolysis was prepared following the Tryptic Digestion Kits (Thermo, USA). Using the EASY-nLCTM 1200 chromatography (Thermo, USA) and Q.Exactive HF-XTM Mass Spectrometry (Thermo,

USA) for protein sequencing analysis. For 16S rDNA high-throughput sequencing, the region V3-V4 of bacterial 16S rDNA gene was amplified. The PCR products were paired-end sequenced through an Illumina MiSeq PE300 platform/NovaSeq PE250 platform.

2.7. RT-PCR amplification of *nim* gene

The *nim* gene in the total DNA of activated sludge was amplified using the general primers NIM-3: (5'-ATGTTTCAGAGAAATGCGGC GTAAGCG-3') and NIM-5: (5'-GCTTCCTTGCTGTGCATGTGCTC-3') through quantitative real-time PCR (q-PCR) (Alauzet et al., 2019). Reactions were performed within QuantStudio® 5 Real-Time PCR Detection System (Thermo ABI, America) with the use of SYBR® Premix Ex TaqTM GC (TAKARA, Japan). The relative expression of *nim* was standardized with the 16S rRNA gene (Yu et al., 2016).

2.8. Analytical methods

The surface structures of polyurethane foam, mycelia of strain LZ-M, and activated sludge were observed using scanning electron microscopy (JEOL JSM-IT500LA). The analysis of COD, total nitrogen, and mixed liquor suspended solids (MLSS) of AAS was conducted following the standard procedure (Carranzo, 2012). Reactive oxygen species (ROS) were obtained through a ROS assay kit (Beyotime Biotechnology, Jiangsu, China). Loosely bound extracellular polymeric substances (LB-EPS) and tightly bound EPS (TB-EPS) were drawn and examined following the method described previously (Wang et al., 2013). Protein was examined by Pierce BCA Protein Assay Kit (Thermo Fisher Scientific), and polysaccharide was obtained through the colorimetric method (Dubois et al., 1956).

Waters Acquity I-class UPLC and a Waters Synapt G2-Si high-definition MS (HDMS) ESI/Q-TOF system was used for the products analysis of MNZ degradation by Wuhan Metware Biotechnology Co., Ltd (Hubei, China). The concentrations of MNZ, FZD and DNSA were obtained through high-performance liquid chromatography (HPLC, Jasco, Japan) at the wavelength of 320 nm, 365 nm, and 304 nm, respectively. An Agilent Eclipse XDB-C18 (4.6 mm × 250 mm, 5 µm) was used, and the mobile phase for MNZ and FZD detection comprised 20% methanol and 30% methanol, respectively. The mobile phase of DNSA detection comprised methanol and 2% acetic acid (50:50). The CO₂ and CH₄ contents of biogas were evaluated using gas chromatography (Agilent 6890 N) with a thermal conductivity detector.

3. Results and discussion

3.1. Degradation of MNZ by *A. tabacinus* LZ-M

The pellets of strain LZ-M were employed to degrade MNZ. It degraded 5 mg/L of MNZ with 75.02 ± 0.82% removal efficiency at 120 h (Fig. 1A). Antibacterial experiments showed that the colony number was close to that of the control without MNZ when the *C. perfringens* solution was treated with MNZ degradation solution. In contrast, the colony number of bacterial solution after being treated with MNZ solution was significantly reduced compared to the control (Fig. 1B). The results illustrated that the MNZ in the medium was degraded by strain LZ-M and the degradation metabolites did not inhibit *C. perfringens*. The biomass concentration of LZ-M affected the degradation of MNZ, and the efficient biomass concentration was above 3 g/L (Fig. 1C). Strain LZ-M can completely degrade MNZ at the concentration of 1 mg/L, while the removal efficiency was decreased to 77.39 ± 2.91%, 27.21 ± 1.18% and 21.79 ± 3.72% at the concentration of 5, 10 and 20 mg/L, respectively (Fig. 1D). The pH test showed that MNZ was removed effectively (63–78%) by LZ-M at pH 5–9 (Fig. 1E). The optimal temperature for MNZ degradation was 30 °C (Fig. 1F). Metronidazole belongs to a nitrogroup of drugs. Under anaerobic conditions, the degradation potential of strain LZ-M on other nitro-drugs, including FZD and DNSA was

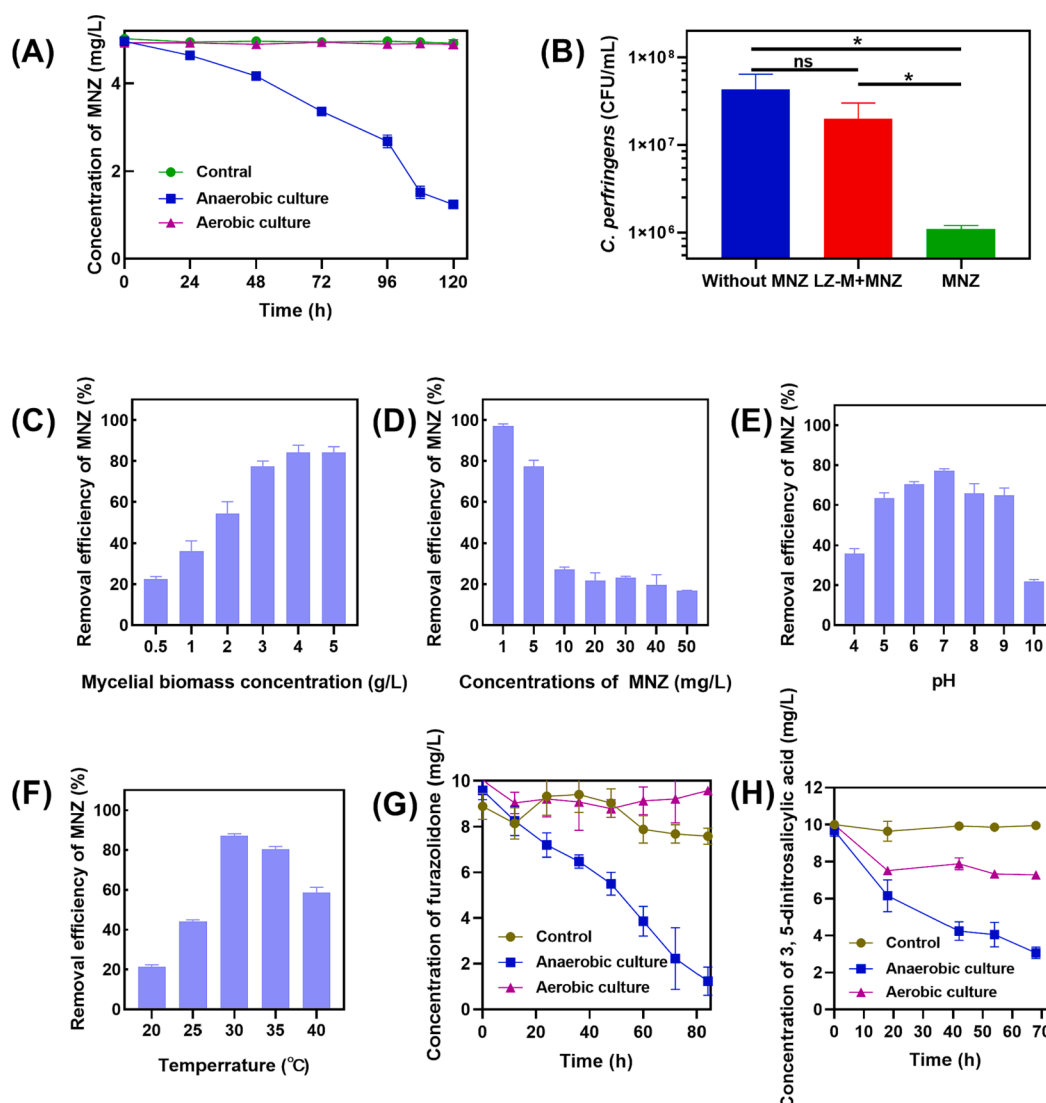


Fig. 1. Anaerobic degradation of MNZ by *A. tabacinus* LZ-M. (A) Variation curve of MNZ concentration with time under aerobic and anaerobic conditions; (B) Bioassay for antibacterial activity of metronidazole metabolites on *Clostridium perfringens*; (C–F) The effect of mycelial biomass concentration, MNZ concentration, pH and temperature on the degradation of MNZ; (G and H) Anaerobic degradation of furazolidone (FZD) and 3,5-dinitrosalicylic acid (DNSA) by *A. tabacinus* LZ-M.

evaluated (Fig. 1G and H). The findings revealed that strain LZ-M degraded $87.03 \pm 7.02\%$ of 10 mg/L FZD and $68.31 \pm 2.20\%$ of 10 mg/L DNSA within 84 h, respectively. The above results suggest that strain LZ-M degrades MNZ, FZD, and DNSA significantly.

In previous studies, filamentous fungi were mostly found to degrade organic substances under aerobic conditions; for example, *Trametes versicolor* and *Panus tigrinus* degrade the antibiotics norfloxacin and ciprofloxacin aerobically (Čvančarová et al., 2015). However, when transferred to an anaerobic environment after aerobic growth, *A. tabacinus* LZ-M has an excellent ability to degrade organic matter. Its organic metabolic process and protein expression differ dramatically under anaerobic conditions, as investigated in a manuscript for publication that examines its anaerobic mechanism. Few microorganisms were found to be capable of degrading furazolidone and metronidazole. Only a few bacteria have been found to degrade 5 mg/L furazolidone, including *Pseudomonas aeruginosa*, *Acinetobacter calcoaceticus*, *Pseudomonas putida*, and *Proteus mirabilis* (Zhang et al., 2013). FZD and DNSA are nitrocompounds commonly used in medicine and pharmaceutical industry, respectively (Zhang et al., 2013; Zhao et al., 2020). The bioremediation of MNZ, FZD, and DNSA illustrated the variety of nitrodrugs that LZ-M can degrade, indicating that *A. tabacinus* LZ-M serves as

a potential microbial resource for bioremediation of the above pollutants.

3.2. Pathway and mechanism of MNZ degradation

LC-MS was used to examine the MNZ degradation products, and seven intermediate metabolites were observed (see supplementary material). The degradation pathway of MNZ was inferred as Fig. 2A. The methanol group of MNZ was removed in the carbon chain via —C—C— bond cleavage and then demethylated by —C—N— bond cleavage to produce 2-methyl-5-nitroimidazole. The above process was also identified in a previous study of MNZ degradation (Pan et al., 2019; Zhang et al., 2022a). The imidazole ring was hydroxylated, followed by —C—N— bond cleavage to open the imidazole ring, and then dissociation to acetylene and urea by —C—N— bond cleavage. The concentration of 2-methyl-5-nitroimidazole reached 0.329 mg/L at 72 h, and urea concentration was 0.269 mg/L at 120 h. (see supplementary material). The results suggest that MNZ is metabolized to urea by strain LZ-M through —C—C— bond cleavage, hydroxylation, and —C—N— bond cleavage.

The gene expression and protein expression of strain LZ-M cultured

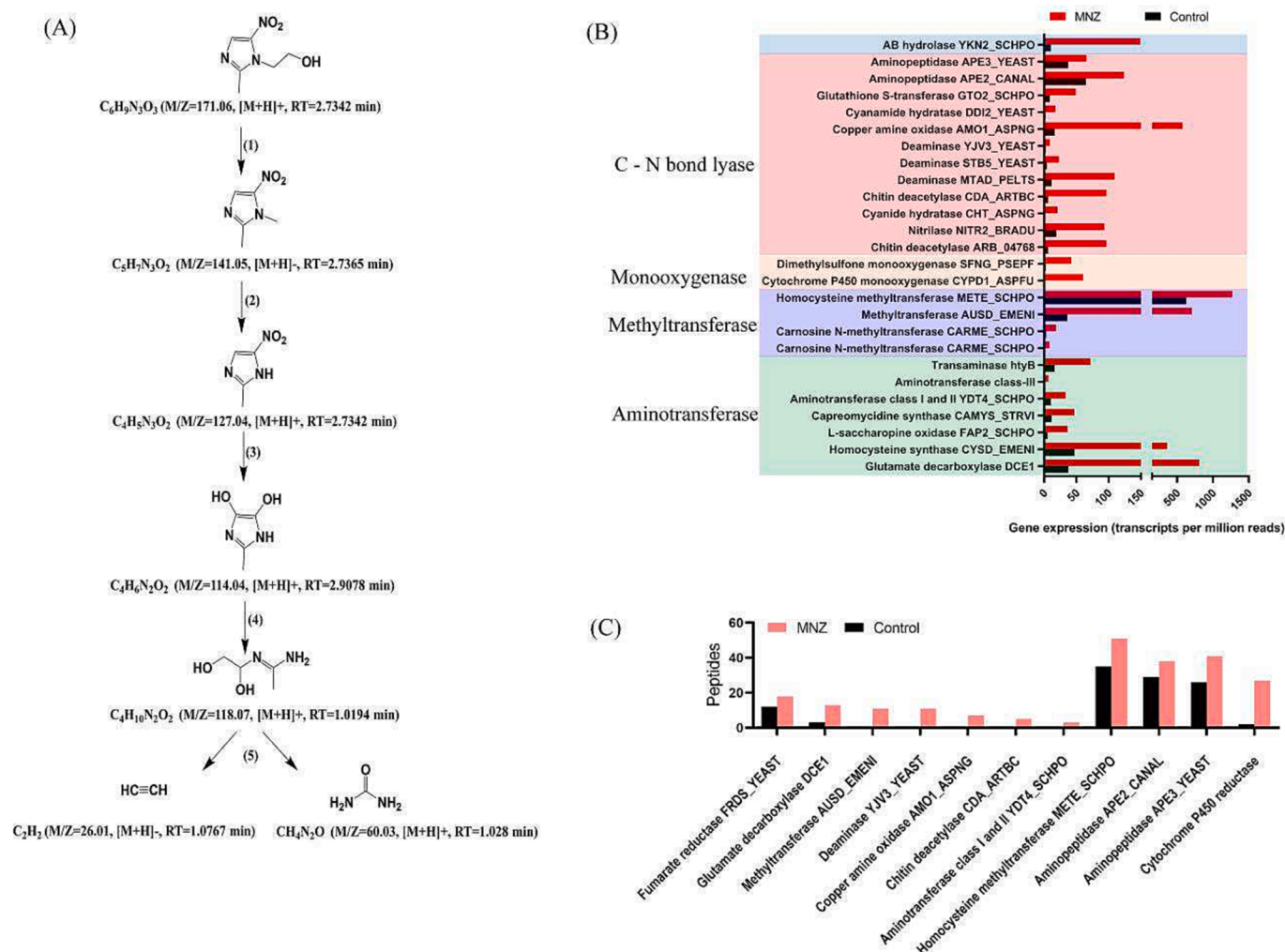


Fig. 2. (A) Putative degradation pathways of metronidazole in *A. tabacinus* LZ-M; (B) Expression of gene related with MNZ degradation in *A. tabacinus* LZ-M by transcriptome sequencing. (C) Number of peptides related to MNZ degradation in *A. tabacinus* LZ-M by proteome sequencing.

with MNZ was investigated in accordance with the transcriptome and proteome. There was a significant difference between a total of 3,867 transcripts in expression under the influence of MNZ, including 2437 transcripts that were up-regulated and 1432 down-regulated transcripts. The metabolic-pathway analysis indicated that the —C—C— cleavage, —C—N— cleavage, and hydroxylation in the process of MNZ degradation. The transcriptome analysis showed that the gene expression of aminotransferase, methyltransferase, monooxygenase, and —C—N— cleavage hydrolase was up-regulated by MNZ culture (Fig. 2B). An existing study states that hydrolase is involved in the cleavage of the side chain acetyl group during the degradation of antibiotic 2,4-Diacetylphloroglucinol (Bottiglieri & Keel, 2006). AB hydrolase reacts on —C—C— bond cleavage (Schleberger et al., 2007), which may remove the methanol group at the carbon chain of MNZ. The second step of MNZ degradation was nitrogen-terminal demethylation by —C—N— bond cleavage, which may be correlated with methyltransferase. Four methyltransferase genes were significantly up-regulated, including cysteine methyltransferase METE, methyltransferase AU5D, and carnosine N-methyltransferase CARME. Carnosine N-methyltransferase has been reported to transfer the N-linked methyl group (Liu et al., 2017). Cytochrome P450 monooxygenase and dimethylsulfone monooxygenase can achieve hydroxylation of unsaturated carbons (Lu et al., 2018), and they may react on the imidazole hydroxylation at the third step. Previous research has also identified that cytochrome P450 monooxygenase in *Chlorella sorokiniana* is involved in the degradation of antibiotic

sulfamethoxazole through H-abstraction-OH-rebound reaction (Chu et al., 2022). Several classes of —C—N— lyase, including aminopeptidase, glutathione S-transferase, deacetylase, deaminase, and amidohydrolase were up-regulated expression significantly, and the above enzymes can break —C—N— bond; for instance, glutathione S-transferase has been found to degrade tetracycline, sulfathiazole, and ampicillin (Park & Choung, 2007). The above enzymes may facilitate the cleavage of the —C—N— bond of MNZ, thus opening the imidazole ring and producing urea. Therefore, the above up-regulated enzymes might be involved in MNZ degradation.

As the result of proteome sequencing (Fig. 2C), the presence of enzymes related to MNZ degradation was found, including cytochrome P450, aminopeptidase, methyltransferase, aminotransferase, copper ammonia oxidoreductase and deaminase. The above enzymes were expressed and up-regulated under the influence of MNZ, which was consistent with the transcriptome result, indicating that they might be involved in MNZ degradation under anaerobic condition.

3.3. Bioaugmentation using immobilized strain LZ-M for treatment of municipal wastewater containing MNZ

The degradation efficiency of four UASB reactors for municipal wastewater treatment was detected. The COD of the effluent of AS, ME, ME+LZ-M, and ME+ImLZ-M reactors reached 465, 883, 317, and 162 mg/L on day 31, and the removal efficiency was 49.46%, 11.7%,

67.28%, and 83.31%, respectively (Fig. 3A). Under the influence of immobilized LZ-M, the removal efficiency of COD was improved by 7.12 times. The immobilized LZ-M outperforms the free LZ-M in bio-augmentation. MNZ was not determined in the effluent of reactors ME+LZ-M and ME+ImLZ-M, while it was detectable for 13 days in the ME reactor (Fig. 3B). The cumulative gas production was examined every day. The total volumes of gas were 105, 63.5, 162.5, and 390 mL in AS, ME, ME+LZ-M, and ME+ImLZ-M systems on day 31, respectively (Fig. 3C). The result showed that MNZ inhibited the gas production of AAS system, and bioaugmentation facilitated the removal of MNZ in wastewater and increased the transformation of organic carbon into biogas. GC analyzed the biogas component; CO₂ was the main component, and its content exceeded 80% in all reactors. ME+ImLZ-M reactor had a highest CH₄ content of 19.6% on the 31st day (see [supplementary material](#)). The possible reason for low CH₄ and high CO₂ in the reactors is that the influent carrying oxygen make it difficult to form strict anaerobic conditions for methanogenesis. The nitrogen content in the above systems fluctuated over time, and it was close in the four systems at the end of the operation (Fig. 3D). The above results indicated that immobilized LZ-M eliminated the inhibitory effect of MNZ on the AAS system and significantly increased the efficiency of wastewater

treatment.

Existing researches have suggested that the accumulation of antibiotics in wastewater reduces the treatment efficiency of bioreactors (Czatkowska et al., 2021; Zhou et al., 2021). In an anaerobic sludge digester, MNZ reduces the system's methane production from 272.8 L/kg VS to 44.3 L/kg VS and 0.26 g/g MNZ residue is still detected in the sludge after 40 days of fermentation (Czatkowska et al., 2021). The anaerobic digestion process in this study did not produce the same concentrations of methane as previously reported due to wastewater replacement. However, both gas suppression and metronidazole residue indicate that the metronidazole's degradation is limited in activated sludge systems. Different fungal strategies have been adopted to eliminate antibiotics in bioreactors. For instance, in a whole-cell immobilization system, the *Penicillium chrysogenum* degrades 90% of tetracyclines (Wu et al., 2021). And *Trametes versicolor* and *Bjerkandera adusta* in combination with wheat straw degrade 85–94% of Sulfamethoxazole, Erythromycin, and Tetracycline (Aydin, 2016). In a fungal biomass fluidize bioreactor, *Trametes versicolor* degrades 98.5 % of ofloxacin in hospital wastewater (Gros et al., 2014). However, the fungal biomass method leads to the rupture of pellets (Espinosa-Ortiz et al., 2016), and the fungus' function is rendered ineffective by whole-cell immobilization (Wu et al., 2021). Porous foams have protective effects on introduced microorganisms (Chen et al., 2022). The immobilized fungus used in this study can degrade 100% of MNZ in wastewater, which is comparable or superior to the above methods. Immobilized bio-augmentation increases the degradation efficiency of organic matter, thus suggesting that the AAS system is protected even when antibiotic contamination is present.

3.4. The effect of bioaugmentation on the properties of activated sludge

The immobilized strain LZ-M was identified by an electron microscope (see [supplementary material](#)). Strain LZ-M exhibited a filamentous structure. After immobilization, its hyphae were attached to the foam's pores. The immobilized LZ-M was cultured in activated sludge reactors for 31 days, and the hyphae and activated sludge were identified in the foam's pores. In some pores, the coexistence of mycelium and activated sludge was discovered. This phenomenon indicated that filamentous fungus LZ-M could coexist with the native microorganisms with the help of immobilization. Polyurethane foam is confirmed as an excellent carrier for the immobilization of filamentous fungi. The biomass concentration of activated sludge was examined, and it increased with time in the ME+ImLZ-M reactor and arrived at 4.09 g/L on day 31, while it was 3.18, 2.03, and 3.07 g/L in AS, ME, and ME+LZ-M reactors, respectively (Fig. 4A). This showed that MNZ inhibited microorganisms' growth in AAS. The ROS value of activated sludge was significantly increased by MNZ in the ME reactor, and it increased to 79 on 3th day. In AS sludge, the value was 33 (Fig. 4B). It indicated that MNZ had a toxic effect on AAS. The extracellular polymer analysis of AAS indicated that its TB-EPS was 49.73, 67.41, 50.84 and 53.28 mg/g VSS in AS, ME, ME+LZ-M and ME+ImLZ-M reactors respectively, and LB-EPS were 19.34, 30.39, 19.47 and 18.78 mg/g VSS (Fig. 4C and D). The EPS layer on the sludge surface may absorb toxic substances, restricting them from entering microbes' interiors and lowering their toxicity to microbes (Yu et al., 2019). This result showed that MNZ stimulated activated sludge to secrete extracellular polymer, damaging the sludge structure. Several antibiotics, including ampicillin, ofloxacin, sulfadiazine, and tetracycline in wastewater, are also reported to stimulate the EPS formation of activated sludge in various bio-systems (Yu et al., 2019; Zhang et al., 2022a). The increase of EPS and ROS of activated sludge caused by MNZ showed the damage of activated sludge. Carriers can protect introduced microorganisms from mechanical damage and outflow, and provide a stable environment for microbial survival in the activated sludge system (Yu et al., 2020). The above results indicate that immobilized bio-augmentation protects the AAS from MNZ and increases its resistibility against toxicants.

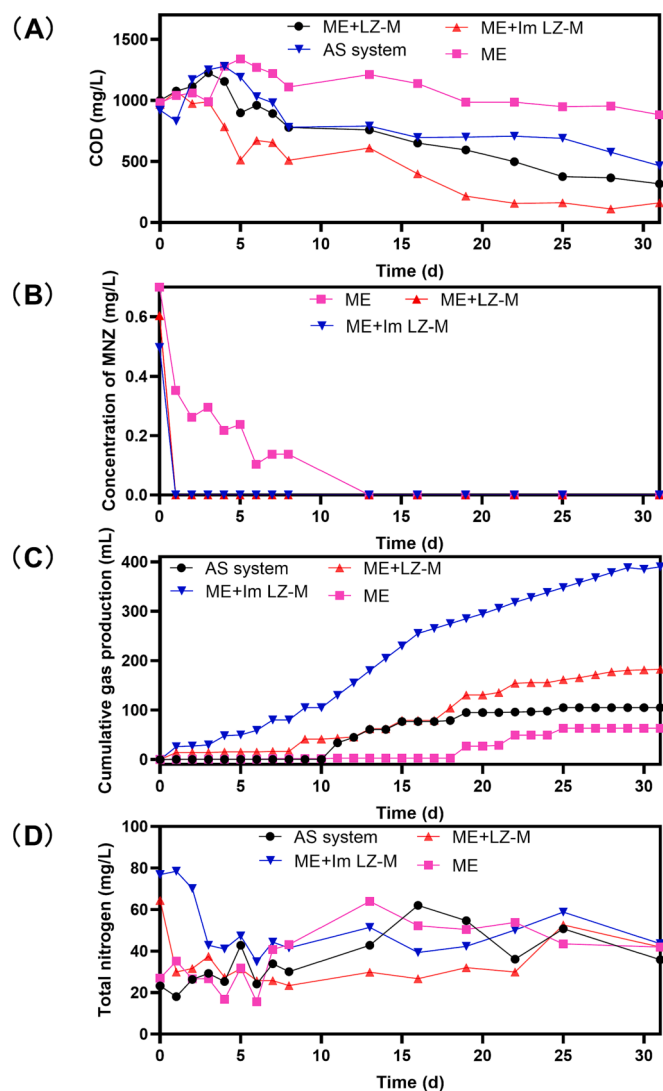


Fig. 3. Treatment efficiencies of municipal wastewater by four AAS reactors. (A) COD concentration of effluent in four reactors; (B) MNZ concentration of effluent in four reactors; (C) Cumulative gas production in four reactors; (D) Concentration of total nitrogen in the effluent of four reactors.

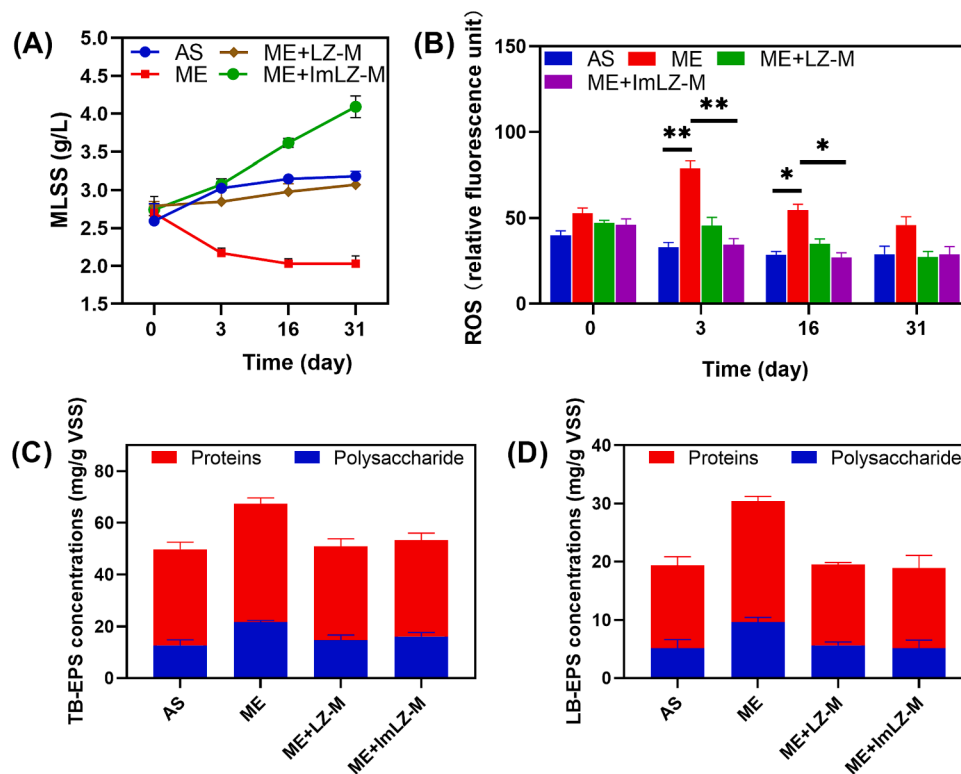


Fig. 4. (A) Concentration of mixed liquor suspended solids (MLSS) of activated sludge in different reactors; (B) The value of ROS of activated sludge in different reactors; (C and D) The concentration and components of loosely bound EPS (LB-EPS) and tightly bound EPS (TB-EPS) of activated sludge in different reactors.

3.5. Microbial community analysis of activated sludge in reactors

The microbial community of activated sludge in the four reactors was studied through the sequencing of 16S rDNA genes (Fig. 5). The result showed that the Shannon index of LZ-M sludge and IM LZ-M sludge was significantly increased compared with that of ME sludge (Fig. 5A). It indicated that the addition of fungus LZ-M improved the diversity of microorganisms. The result of the Ace index indicated that the richness of activated sludge was reduced by MNZ in ME reactor compared with that of AS sludge (Fig. 5B). The result of the cluster analysis indicated that the community structures of ME sludge and AS sludge were significantly different at the PC1 level, whereas those of the LZ-M sludge, the IM LZ-M sludge, and the AS sludge were close at the PC1 level (Fig. 5C). The above result suggests that MNZ significantly affects the community structure of AAS, while its effect was reduced by bioaugmentation. The phylum-level community structure indicated that the microbial communities of ME sludge were different from those of other groups, primarily manifested as the increase of the abundance of Spirochaetota, Desulfobacterota, Proteobacteria, as well as the reduction of the abundance of Bacteroidetes, Patescibacteria, and Chloroflexi (Fig. 5D). The genus-level analysis indicated that when compared with AS sludge, MNZ significantly reduced the abundance of *norank_f_Saprospiraceae*, *norank_c_OLB14*, and *norank_f_Vicinamibacteraceae* from 8.84% to 2.00%, from 2.12% to 0.26%, as well as from 0.82% to 0.13%, respectively, while it enhanced the abundance of *norank_o_Aminicenanales* from 1.43% to 4.70% (Fig. 5E and F). Immobilized bioaugmentation reduced the effect of MNZ on the above genera, such that the abundance of *norank_f_Saprospiraceae*, *norank_c_OLB14*, and *norank_f_Vicinamibacteraceae* was enhanced to 6.94%, 3.67%, and 1.23%, respectively, whereas the abundance of *norank_o_Aminicenanales* was reduced to 0.75%. The above results suggest that the method of immobilized bioaugmentation effectively reduces MNZ' toxic on the microorganisms of ASS while increasing the stability of community structure in systems.

A previous study has also found that MNZ increases the abundance of Proteobacteria and decreases the abundance of Bacteroidetes and Chloroflexi in an anaerobic fermentation system (Czatzkowska et al., 2021). MNZ was used to treat anaerobic bacteria, containing many species belonging to Bacteroidetes, i.e., one of the major bacterial phyla playing a certain role in anaerobic fermentation (Czatzkowska et al., 2021). Both previous study and this study suggest that Bacteroidetes are sensitive to MNZ contamination in AAS systems, and the decline of gas production and COD removal in the AAS systems may be correlated with the decrease of Bacteroidetes (Czatzkowska et al., 2021). The genera reduced by MNZ primarily include *Saprospiraceae* (within the phylum Bacteroidetes), *Vicinamibacteraceae* (within the phylum Acidobacteria), and *OLB14* (within the phylum Chloroflexi). The *OLB14* has been commonly found in biological treatment systems to remove nitrogen and refractory organic compounds (Wang et al., 2019; Zhao et al., 2021). *Saprospiraceae* have been commonly identified in activated sludge wastewater treatment systems. Its family members take on a critical significance in the decomposition of complicated organics (McIlroy & Nielsen, 2014). Existing research has suggested that some *Saprospiraceae* species are capable utilizing and hydrolyzing complicated carbon sources (McIlroy & Nielsen, 2014). The above microorganisms that MNZ has inhibited may function in the organics decomposition during the treatment of municipal wastewater. Immobilized bioaugmentation can protect the above microorganisms from MNZ inhibition, thus suggesting that filamentous fungi benefit the community's stability in AAS system so that the efficient treatment of municipal wastewater. In addition, MNZ increases the abundance of *Aminicenanales* in sludge. *Aminicenanales* have been commonly identified in anaerobic reactors, and they are correlated with the detoxification of several toxic substances (e.g., phenols and *N*-heterocyclic compounds) (Fan et al., 2022). Accordingly, *Aminicenanales* may be correlated with the detoxification of MNZ. The microbial community structure is not affected by toxic substances when bioaugmentation facilitates the degradation of hazardous compounds (Hailei et al., 2017). Bioaugmentation significantly reduces MNZ-

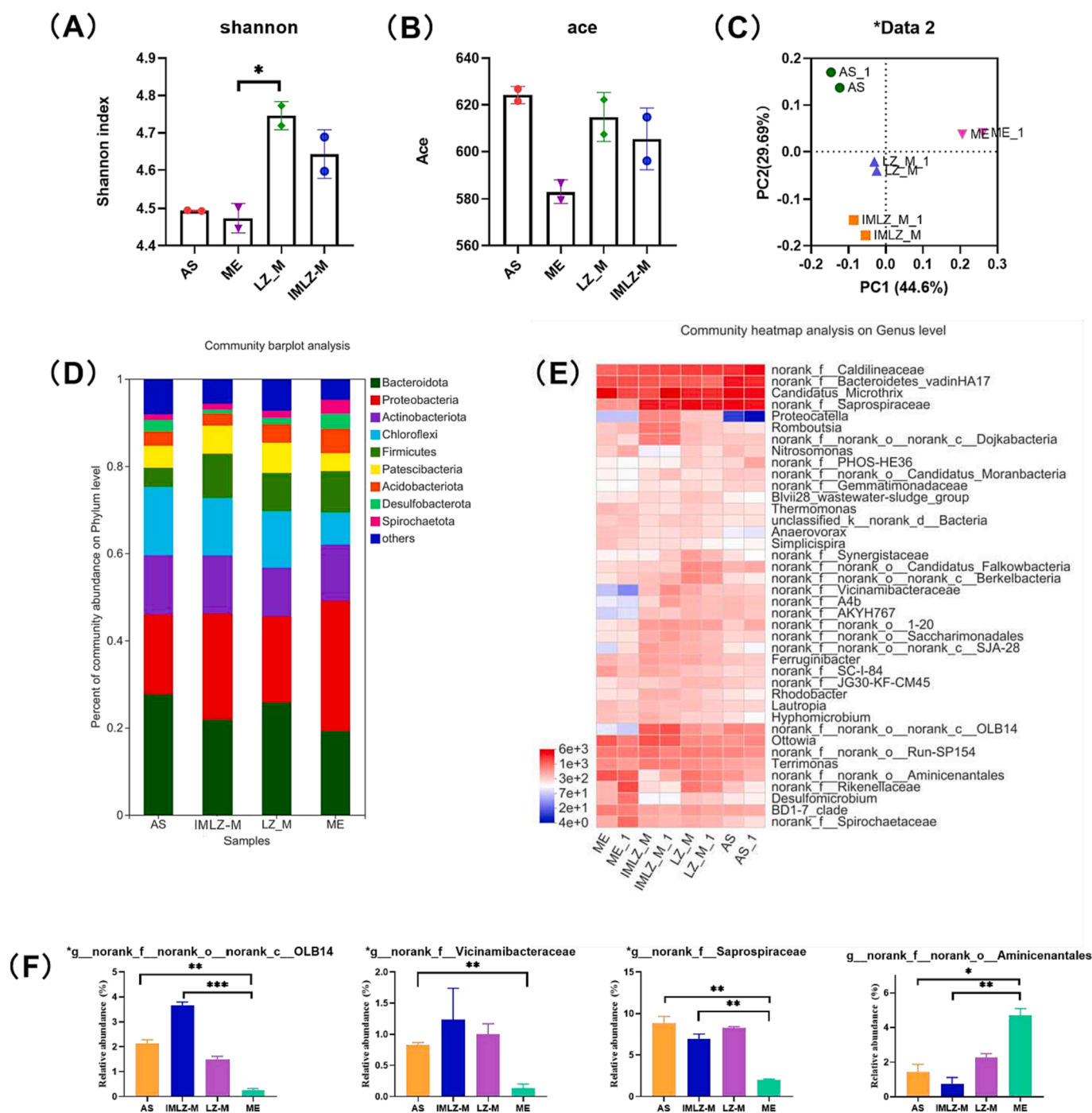


Fig. 5. Microbial community of activated sludge in different reactors. (A–C) Shannon index, Ace index, and principal coordinate analysis of microbial community; (D and E) microbial components at the phylum and genus level; (F) relative abundance of four genera in activated sludge of four reactors. LZ_M sludge and IMLZ_M sludge were from ME+LZ-M and ME+ImLZ-M reactors, respectively (mean $\pm$ SE, $n = 2$).

induced changes in microbial community structure, possibly due to MNZ degradation by LZ-M.

3.6. The effect of bioaugmentation on the metabolic function and resistance gene of activated sludge

The function prediction was performed by PICRUSt2 in accordance with the results of 16S rDNA sequencing (Fig. 6A). Compared with AS sludge, MNZ significantly reduced the metabolism functions of microorganisms in ME sludge, including the metabolisms of nucleotide, lipid, glycan, energy, carbohydrate, and amino acid. However, the above

metabolisms were significantly increased through immobilized bioaugmentation in IMLZ-M sludge, especially carbohydrate metabolism and amino acid metabolism. The metabolism of amino acids allows the synthesis of proteins and polypeptides, and these compounds take on a critical significance in microbial growth and proliferation (Hua et al., 2020). Lipid metabolism is likely to be correlated with the formation of cell membrane structure (de Andrade Melo-Sterza & Poehland, 2021). Proteins involved in carbohydrate metabolism may enhance organics breakdown (Hua et al., 2020). The increase of microbial metabolism in the ME+IMLZ-M reactor explained the more efficient COD removal and the higher biomass concentration of activated sludge. In addition,

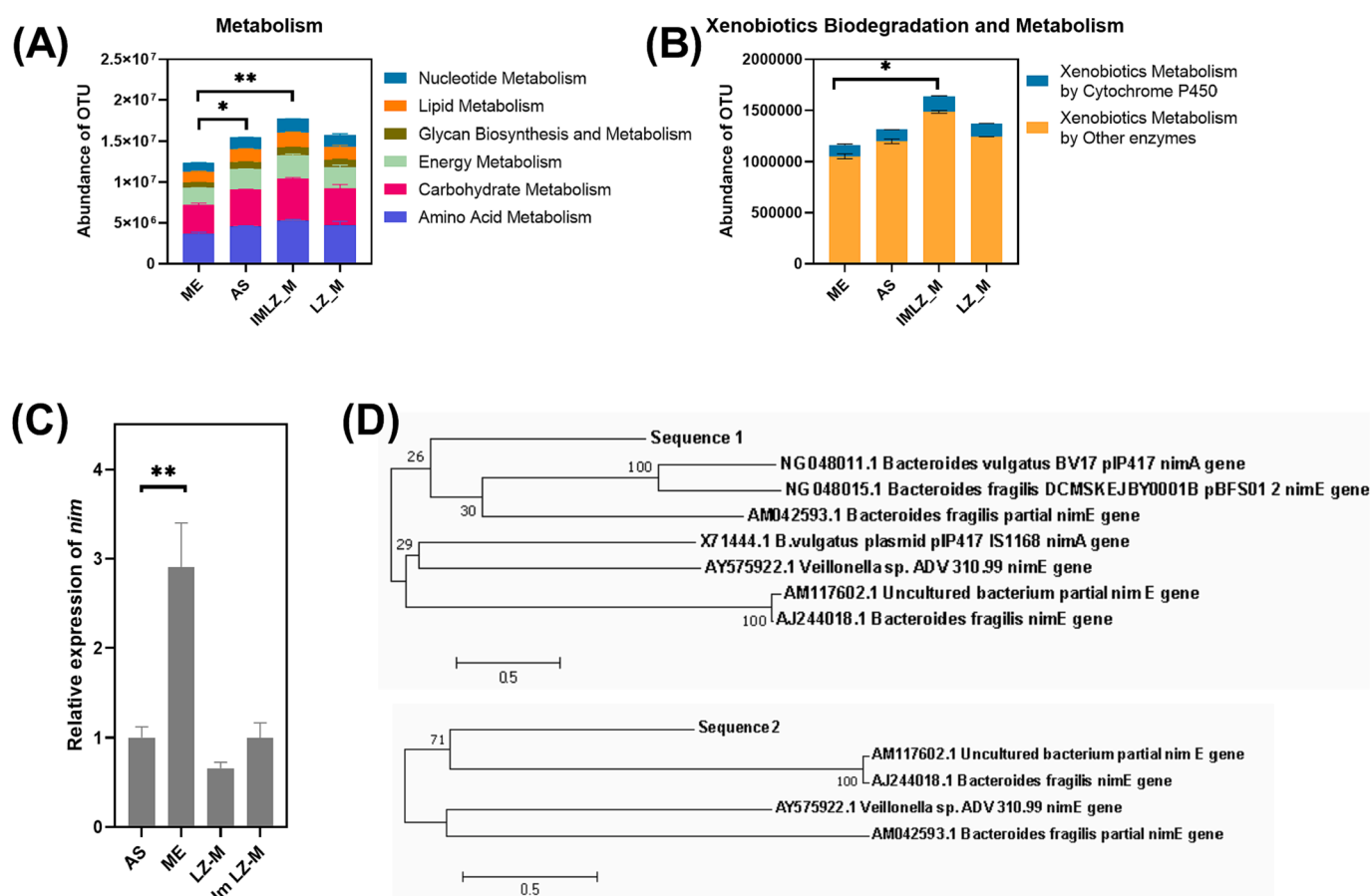


Fig. 6. (A and B) the function prediction of activated sludge by PICRUST2 based on the results of 16S rDNA sequencing data; (C) Relative abundance of *nim* genes in activated sludge determined by RT-PCR; (D) Sequencing alignment results of amplified products in NCBI database.

immobilized bioaugmentation significantly increased the biodegradation and metabolism of xenobiotics, including xenobiotics metabolism by cytochrome P450 and other enzymes (Fig. 6B). It indicated that the degradation of toxic substances was facilitated by bioaugmentation. Thus, bioaugmentation not only protects the activated sludge from antibiotics, but also improves its ability for the degradation of these toxic substances. The functional alteration may be illustrated by the changes of microbial composition in reactors (Fan et al., 2022). After bioaugmentation, the microbial community is excellent and efficient for wastewater treatment.

Nim genes encoding nitro-imidazole-reductases are the main MNZ resistance genes (Alauzet et al., 2019). Trinh and Reyssset proposed a PCR approach using general primers NIM3 and NIM5 that can identify all subtypes of *nim* genes through targeting conservative sequence, and it has been extensively adopted to investigate the *nim* genes in anaerobes thus far (Alauzet et al., 2019). As depicted in the gel electrophoresis image of *nim* gene amplification, two gene bands were generated in different sludge samples, and the product bands of ME sludge were brighter than other types of sludge (see supplementary material). The RT-PCR result indicated that the abundance of *nim* in ME sludge was 2.9 times that of the AS group, whereas the induction of MNZ was inhibited after bioaugmentation (Fig. 6C). The amplified products were sequenced, and the sequences showed an overlapping region with *nim*-related gene sequences (i.e., *nimA* and *nimE*) (Fig. 6D). The above result suggests that MNZ up-regulates the expression of *nim* gene in AS system, and bioaugmentation by immobilized LZ-M is efficient in decreasing the resistance gene contamination caused by MNZ in bioreactors. Existing research has suggested that the abundance of ARGs in compost is reduced by bioaugmentation (Zhong et al., 2021). This study demonstrates that immobilization of filamentous fungi is an effective method to

protect the AAS systems in the presence of MNZ contamination, while reducing the contamination of resistance genes that is caused by antibiotics.

A small concentration of antibiotics in municipal wastewater can cause resistance gene propagation, and activated sludge contains a large number of resistance genes in a wide variety (Zhang et al., 2022b). There are, however, limited ways to lessen the impact of antibiotics on the activated sludge system. This study suggests the effectiveness of immobilized fungi in solving this problem. The fungus strain LZ-M is an excellent antibiotic degrader, capable of degrading MNZ and furazolidone. By polyurethane foam immobilization technology, fungi can coexist with native microorganisms in activated sludge, and the stability and degradation ability of fungi to antibiotics are improved. The results show that the immobilized fungus can effectively protect the sludge system and increase the efficiency of COD removal in municipal wastewater. This method combines the action of fungi and other microorganisms, and the combined degradation of microorganisms has great application expansibility. Simultaneously, this method does not require the pretreatment of urban wastewater during the application process. It has the advantages of low cost, simple application, and good feasibility over the improved methods of adding enzymes (Kumar et al., 2019) or using special reactors (Aydin, 2016; Gros et al., 2014). The immobilized fungus is effective at treating metronidazole in municipal wastewater, but its application to other antibiotics requires additional research. Furthermore, this study is a laboratory-scale experiment that requires additional testing and optimization before large-scale application in real industry.

4. Conclusion

Aspergillus tabacinus LZ-M was capable of degrading 5 mg/L of MNZ under anaerobic conditions. The degradation products showed that MNZ was metabolized to urea. Immobilized LZ-M was introduced into AAS systems for bioaugmentation, effectively removing MNZ from wastewater and increasing COD degradation efficiency and gas production. Furthermore, bioaugmentation reduced the toxic effect of MNZ on activated sludge, facilitated its metabolism and growth, and maintained microbial community stability. MNZ in the AAS system caused resistance gene contamination, which was reduced by bioaugmentation. Effective methods were determined to protect activated sludge from the antibiotic inhibition effect.

CRediT authorship contribution statement

Xuan Yu: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft. **Chunlan Mao:** Project administration, Visualization, Writing – review & editing. **Wenxue Wang:** Methodology, Data curation. **Saurabh Kulshrestha:** Writing – review & editing. **Peng Zhang:** Funding acquisition, Resources. **Muhammad Usman:** Writing – review & editing. **Simin Zong:** Investigation, Methodology. **Mian Gul Hilal:** Writing – review & editing. **Yitian Fang:** Writing – review & editing. **Huawen Han:** Writing – review & editing. **Xiangkai Li:** Conceptualization, Funding acquisition, Project administration, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to my data at the Attach File step

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Date availability

Transcriptome data of *A. tabacinus* LZ-M cultured in the presence or absence of MNZ is stored at Sequence Read Archive (SRA, <https://www.ncbi.nlm.nih.gov/>) under the numbers SRR14455359 and SRR14455362, respectively. The ITS sequence of *A. tabacinus* LZ-M and the original data of 16S rDNA sequencing of activated sludge is stored at SRA database under the number MZ127527 and PRJNA868866.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2022.128509>.

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